

Order Hemiptera *continued*

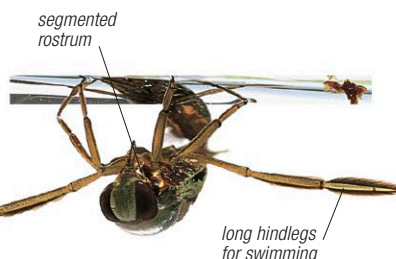
Family Notonectidae

Back-swimmers

Occurrence 350 spp. worldwide; in still water, mostly small pools, ponds, edges of lakes



These spindle-shaped bugs, $\frac{1}{16}$ – $\frac{1}{2}$ in (2–17 mm) long, are good fliers, but spend much of their lives underwater, swimming upside down. They use their forelegs to catch prey and their hindlegs for swimming.

**NOTONECTA GLAUCA**

Like other back-swimmers, this European species stores air under its wings. It detects its prey by sight, and by sensing surface ripples.

Family Pentatomidae

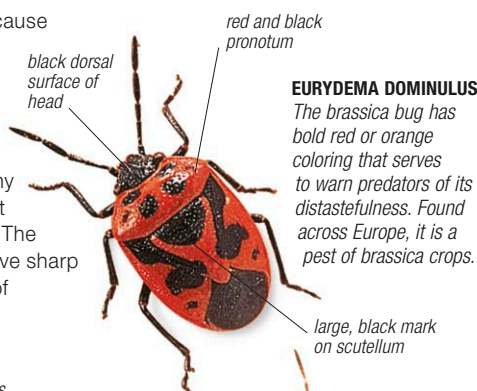
Stink bugs

Occurrence 5,500 spp. worldwide; on herbaceous plants, shrubs, trees

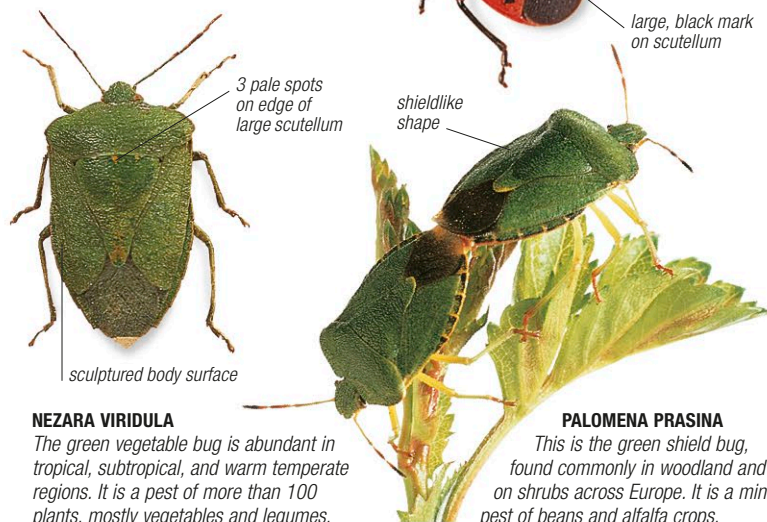


Also known as shield bugs because of their shieldlike shape, these bugs are $\frac{3}{16}$ –1 in (0.5–2.5 cm) long. The name stink bugs comes from their ability to give off strong defensive odors from their thoracic glands. Many species are green or brown but some may be brightly colored. The pronotum is broad and may have sharp corners. Females lay clusters of

barrel-shaped eggs, up to 400 at a time, on plants. There are 5 nymphal stages. Nymphs are mostly herbivorous to begin with but some turn predatory later, or become mixed feeders. Several species are serious pests of a wide-ranging variety of crops, such as legumes, potato, cotton, and rice.

**EURYDEMA DOMINULUS**

The brassica bug has bold red or orange coloring that serves to warn predators of its distastefulness. Found across Europe, it is a pest of brassica crops.

**NEZARA VIRIDULA**

The green vegetable bug is abundant in tropical, subtropical, and warm temperate regions. It is a pest of more than 100 plants, mostly vegetables and legumes.

PALOMENA PRASINA

This is the green shield bug, found commonly in woodland and on shrubs across Europe. It is a minor pest of beans and alfalfa crops.

Family Psyllidae

Jumping plant lice

Occurrence 1,500 spp. worldwide; on stems, leaves, and bark of herbaceous and woody shrubs, trees, crops



Looking like small leafhoppers (see p.558) with long antennae, jumping plant lice are $\frac{3}{64}$ – $\frac{3}{16}$ in (1.5–5 mm) long, and have 2 pairs of oval wings, held rooflike over the body. They feed on plants, and some species are pests. The females lay stalked eggs on, or in, food plants.

**CACOPSYLLA PYRICOLA**

Abundant in the Northern Hemisphere, the pear psylla is a major pest of pear trees, from the leaves of which adults and nymphs suck sap.

Family Reduviidae

Assassin bugs

Occurrence 6,000 spp. worldwide, especially tropical and subtropical regions; on ground, in leaf litter, on plants



Assassin bugs get their name from their predatory habits, although some live by sucking blood. Most are $\frac{1}{32}$ – $1\frac{1}{2}$ in (0.7–4 cm) long, with dark, slender bodies. Many attack insect pests; others suck blood of humans—transmitting infection such as Chagas disease.

**PLATYMERIS BIGUTTATA**

This is a large, highly predatory Afrotropical species, $1\frac{1}{4}$ – $1\frac{1}{2}$ in (3–3.6 cm) long. Its toxic saliva may cause temporary blindness.

Family Tingidae

Lace bugs

Occurrence 2,000 spp. worldwide; on foliage of herbaceous plants, trees



Lace bugs are small, grayish insects, recognizable by the complex, lacelike patterns on their wings and upper body. They are $\frac{1}{16}$ – $\frac{3}{16}$ in (2–5 mm) long. The pronotum can extend, hoodlike, over the head. Females of some species take care of the eggs and nymphs. Many are pests on plants.

**TINGIS CARDUI**

The spear-thistle lace bug is common in the UK and eats certain types of thistles. A covering of light, powdery wax gives it a pale gray look.

Order Thysanoptera

Thrips

This order consists of 7,400 species within 14 families. Thrips are small and slender, with 2 pairs of narrow, hair-fringed wings. They have large eyes, short antennae, and sucking mouthparts, with one small and one needlelike mandible. A sticky, inflatable structure between the claws aids grip. Metamorphosis is unusual, with 2 nymphal stages and one or more pupalike stages.

Family Thripidae

Common thrips

Occurrence 2,000 spp. worldwide; in leaf litter, on leaves, flowers, and fruit of a wide range of plants



In warm, humid weather, clouds of common thrips, or thunderbugs, form part of the aerial plankton—myriad tiny, winged insects that are blown far and wide by the breeze. Common thrips vary from pale yellow to brown or black, and have very narrow, feathery wings. Typical species are less than $\frac{1}{16}$ in (2 mm) long. Some common thrips reproduce asexually, and in most, the female inserts eggs into plant tissues or flowers using a tiny, sawlike ovipositor. The nymphs develop either on plants or in the soil. Adults and nymphs generally live by sucking plant juices, although a few attack other insects. Some species are serious pests of tobacco, cotton, beans, and other crops.

**THRIPS FUSCIPENNIS**

This dark-colored species, the rose thrips, with distinctive hair on its body, is abundant in the Northern Hemisphere, and is found on a wide range of plant species.

**THRIPS SIMPLEX**

The gladiolus thrips was originally from South Africa, but is now also widespread in Europe and North America, wherever gladiolus flowers grow.